BCA111P – Lab Sheet 05

Answer Key: Control Structures in C++ | 30 Programs

Each program is standalone. Save each one in a separate .cpp file and compile/run it.

1. Print 1 to 10, skipping 5 using continue

```
#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 5) continue;
        cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

2. Print 1 to 10, stopping at 7 using break

```
#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 7) break;
        cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

3. Search for a number in a sequence using break

```
#include <iostream>
using namespace std;

int main() {
    int numbers[] = {10, 20, 30, 40, 50};
    int target;
    cout << "Enter number to search: ";
    cin >> target;

    bool found = false;
    for (int i = 0; i < 5; i++) {
        if (numbers[i] == target) {
            cout << "Number found at position " << i + 1 << endl;
            found = true;
            break;
        }
    }
    if (!found) cout << "Number not found" << endl;
    return 0;
}
```

4. Even numbers from 1 to 20, skipping multiples of 4

```
#include <iostream>
using namespace std;

int main() {
    for (int i = 2; i <= 20; i += 2) {
        if (i % 4 == 0) continue;
        cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

5. Menu-driven calculator using switch and default

```
#include <iostream>
using namespace std;

int main() {
    double a, b;
    char op;
    cout << "Enter expression (e.g. 5 + 2): ";
    cin >> a >> op >> b;

    switch (op) {
        case '+': cout << a + b; break;
        case '-': cout << a - b; break;
        case '*': cout << a * b; break;
        case '/':
            if (b != 0) cout << a / b;
            else cout << "Cannot divide by zero";
            break;
        default: cout << "Invalid operator";
    }
    cout << endl;
    return 0;
}
```

6. Check prime number and terminate loop early using break

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a positive integer: ";
    cin >> n;

    bool prime = (n >= 2);
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            prime = false;
            break;
        }
    }

    if (prime) cout << "Prime number";
    else cout << "Not a prime number";
    cout << endl;
    return 0;
}
```

7. Read numbers until -1; skip other negative numbers

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter numbers (-1 to stop): ";

    while (cin >> n) {
        if (n == -1) break;
        if (n < 0) continue;
        cout << "Accepted: " << n << endl;
    }
    return 0;
}
```

8. Multiplication table stops when product exceeds 50

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;

    for (int i = 1; ; i++) {
        int product = n * i;
        if (product > 50) break;
        cout << n << " x " << i << " = " << product << endl;
    }
    return 0;
}
```

9. Demonstrate default case in switch

```
#include <iostream>
using namespace std;

int main() {
    int choice;
    cout << "Enter a number from 1 to 3: ";
    cin >> choice;

    switch (choice) {
        case 1: cout << "You selected One"; break;
        case 2: cout << "You selected Two"; break;
        case 3: cout << "You selected Three"; break;
        default: cout << "No matching case";
    }
    cout << endl;
    return 0;
}
```

10. Count positive numbers until 0 is entered

```
#include <iostream>
using namespace std;

int main() {
    int n, count = 0;
    cout << "Enter numbers (0 to stop): ";

    while (cin >> n) {
        if (n == 0) break;
        if (n > 0) count++;
    }
    cout << "Positive numbers entered: " << count << endl;
    return 0;
}
```

11. Factorial using while loop

```
#include <iostream>
using namespace std;

int main() {
    int n;
    unsigned long long factorial = 1;
    cout << "Enter a number (0 to 20): ";
    cin >> n;

    if (n < 0 || n > 20) {
        cout << "Enter a number from 0 to 20";
        return 0;
    }

    int i = 1;
    while (i <= n) {
        factorial *= i;
        i++;
    }
    cout << "Factorial = " << factorial << endl;
    return 0;
}
```

12. Fibonacci series using while loop

```
#include <iostream>
using namespace std;

int main() {
    int terms;
    cout << "Enter number of terms: ";
    cin >> terms;

    long long a = 0, b = 1;
    int count = 0;
    while (count < terms) {
        cout << a << " ";
        long long next = a + b;
        a = b;
        b = next;
        count++;
    }
    cout << endl;
    return 0;
}
```

13. Print all prime numbers between 1 and N

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter N: ";
    cin >> n;

    for (int number = 2; number <= n; number++) {
        bool prime = true;
        for (int d = 2; d * d <= number; d++) {
            if (number % d == 0) {
                prime = false;
                break;
            }
        }
        if (prime) cout << number << " ";
    }
    cout << endl;
    return 0;
}
```

14. Check whether a number is an Armstrong number

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a non-negative integer: ";
    cin >> n;
    if (n < 0) {
        cout << "Not an Armstrong number";
        return 0;
    }

    int temp = n, digits = 0;
    do { digits++; temp /= 10; } while (temp != 0);

    temp = n;
    long long sum = 0;
    do {
        int digit = temp % 10;
        long long power = 1;
        for (int i = 0; i < digits; i++) power *= digit;
        sum += power;
        temp /= 10;
    } while (temp != 0);

    if (sum == n) cout << "Armstrong number";
    else cout << "Not an Armstrong number";
    cout << endl;
    return 0;
}
```

15. Armstrong numbers between 1 and 500

```
#include <iostream>
using namespace std;

int main() {
    for (int n = 1; n <= 500; n++) {
        int temp = n, sum = 0;
        do {
            int digit = temp % 10;
            sum += digit * digit * digit;
            temp /= 10;
        } while (temp != 0);

        if (sum == n) cout << n << " ";
    }
    cout << endl;
    return 0;
}
```

16. Check whether a number is a perfect number

```
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cout << "Enter a positive integer: ";
    cin >> n;

    for (int i = 1; i <= n / 2; i++) {
        if (n % i == 0) sum += i;
    }

    if (n > 1 && sum == n) cout << "Perfect number";
    else cout << "Not a perfect number";
    cout << endl;
    return 0;
}
```

17. Perfect numbers between 1 and 1000

```
#include <iostream>
using namespace std;

int main() {
    for (int n = 2; n <= 1000; n++) {
        int sum = 0;
        for (int i = 1; i <= n / 2; i++) {
            if (n % i == 0) sum += i;
        }
        if (sum == n) cout << n << " ";
    }
    cout << endl;
    return 0;
}
```

18. Check whether a number is a strong number

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a non-negative integer: ";
    cin >> n;
    if (n < 0) {
        cout << "Not a strong number";
        return 0;
    }

    int temp = n, sum = 0;
    do {
        int digit = temp % 10;
        int fact = 1;
        for (int i = 1; i <= digit; i++) fact *= i;
        sum += fact;
        temp /= 10;
    } while (temp != 0);

    if (sum == n) cout << "Strong number";
    else cout << "Not a strong number";
    cout << endl;
    return 0;
}
```

19. Strong numbers between 1 and 500

```
#include <iostream>
using namespace std;

int main() {
    for (int n = 1; n <= 500; n++) {
        int temp = n, sum = 0;
        do {
            int digit = temp % 10;
            int fact = 1;
            for (int i = 1; i <= digit; i++) fact *= i;
            sum += fact;
            temp /= 10;
        } while (temp != 0);

        if (sum == n) cout << n << " ";
    }
    cout << endl;
    return 0;
}
```

20. Reverse a number and check palindrome

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a non-negative integer: ";
    cin >> n;
    if (n < 0) {
        cout << "Please enter a non-negative integer";
        return 0;
    }

    int original = n, reverse = 0;
    do {
        reverse = reverse * 10 + n % 10;
        n /= 10;
    } while (n != 0);

    cout << "Reverse = " << reverse << endl;
    if (original == reverse) cout << "Palindrome number";
    else cout << "Not a palindrome number";
    cout << endl;
    return 0;
}
```

21. Sum of even and odd digits

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a non-negative integer: ";
    cin >> n;
    if (n < 0) {
        cout << "Please enter a non-negative integer";
        return 0;
    }

    int evenSum = 0, oddSum = 0;
    do {
        int digit = n % 10;
        if (digit % 2 == 0) evenSum += digit;
        else oddSum += digit;
        n /= 10;
    } while (n != 0);

    cout << "Sum of even digits = " << evenSum << endl;
    cout << "Sum of odd digits = " << oddSum << endl;
    return 0;
}
```

22. Check whether a number is a Harshad number

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a positive integer: ";
    cin >> n;
    if (n <= 0) {
        cout << "Enter a positive integer";
        return 0;
    }

    int temp = n, digitSum = 0;
    while (temp > 0) {
        digitSum += temp % 10;
        temp /= 10;
    }

    if (n % digitSum == 0) cout << "Harshad number";
    else cout << "Not a Harshad number";
    cout << endl;
    return 0;
}
```

23. Harshad numbers between 1 and 100

```
#include <iostream>
using namespace std;

int main() {
    for (int n = 1; n <= 100; n++) {
        int temp = n, digitSum = 0;
        while (temp > 0) {
            digitSum += temp % 10;
            temp /= 10;
        }
        if (n % digitSum == 0) cout << n << " ";
    }
    cout << endl;
    return 0;
}
```

24. Multiplication tables from 1 to 10

```
#include <iostream>
using namespace std;

int main() {
    for (int n = 1; n <= 10; n++) {
        cout << "Table of " << n << ":" << endl;
        for (int i = 1; i <= 10; i++) {
            cout << n << " x " << i << " = " << n * i << endl;
        }
        cout << endl;
    }
    return 0;
}
```

25. Print prime factors of a number

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter an integer greater than 1: ";
    cin >> n;
    if (n <= 1) {
        cout << "Please enter a number greater than 1";
        return 0;
    }

    cout << "Prime factors: ";
    for (int factor = 2; factor <= n / factor; factor++) {
        while (n % factor == 0) {
            cout << factor << " ";
            n /= factor;
        }
    }
    if (n > 1) cout << n;
    cout << endl;
    return 0;
}
```

26. Find LCM of two positive numbers using loops

```
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cout << "Enter two positive integers: ";
    cin >> a >> b;
    if (a <= 0 || b <= 0) {
        cout << "Both numbers must be positive";
        return 0;
    }

    int larger = (a > b) ? a : b;
    int lcm = larger;
    while (lcm % a != 0 || lcm % b != 0) {
        lcm += larger;
    }
    cout << "LCM = " << lcm << endl;
    return 0;
}
```

27. Find GCD of two positive numbers using loops

```
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cout << "Enter two non-negative integers: ";
    cin >> a >> b;

    int x = a, y = b;
    while (y != 0) {
        int remainder = x % y;
        x = y;
        y = remainder;
    }
    cout << "GCD = " << x << endl;
    return 0;
}
```

28. Sum of series 1 + 2 + 3 + ... + N

```
#include <iostream>
using namespace std;

int main() {
    int n;
    long long sum = 0;
    cout << "Enter N: ";
    cin >> n;

    for (int i = 1; i <= n; i++) sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

29. Sum of squares 1^2 + 2^2 + ... + N^2

```
#include <iostream>
using namespace std;

int main() {
    int n;
    long long sum = 0;
    cout << "Enter N: ";
    cin >> n;

    for (int i = 1; i <= n; i++) sum += 1LL * i * i;
    cout << "Sum of squares = " << sum << endl;
    return 0;
}
```

30. Sum of cubes 1^3 + 2^3 + ... + N^3

```
#include <iostream>
using namespace std;

int main() {
    int n;
    long long sum = 0;
    cout << "Enter N: ";
    cin >> n;

    for (int i = 1; i <= n; i++) sum += 1LL * i * i * i;
    cout << "Sum of cubes = " << sum << endl;
    return 0;
}
```